

Self-Improving AI as an Endogenous Growth System with Volatility-Driven Innovation and Multi-Agent Selection

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Abstract

We study n competing self-improving AI agents, each with algorithmic capital K_i , endogenous volatility $v_i = \sigma_i^2$, and resource share x_i . Volatility raises the rate and size of architectural innovations, capital raises volatility, and shares evolve by replicator dynamics. Using standard results from positive systems, Markov-process stability, and evolutionary game theory, we derive (i) a single loop-gain condition for stability of the capital–volatility block, (ii) a saddle-node transition to runaway growth, (iii) a drift condition giving positive recurrence of the stochastic system, and (iv) a selection result for monopoly. The Hopf bifurcation and the spillover threshold are not derived; the latter is stated as a conjecture.

The paper ends with “The End”

1 Model

Let $K_i \geq 0$, $v_i \geq 0$, and x lie in the simplex. Agent i innovates at rate $\Lambda(v_i)$, with Λ increasing, and each innovation adds capital with mean $j(v_i)$. If candidate architectures have outcome spread $\sqrt{v}$, an option-value argument gives j increasing in v ; for Gaussian outcomes $j(v) = \sqrt{v/2\pi}$. The innovation drift is

$$\psi(v) = \Lambda(v) j(v), \quad \psi(0) = 0, \quad \psi' \geq 0. \quad (1)$$

For the oliGARCH recursion $v_{t+1} = \omega + \alpha \epsilon^2 + \beta v + \gamma \circ K$ with $\epsilon_j = \sqrt{v_j} z_j$ and z_j i.i.d. with mean 0 and variance 1 [2, 3], we have $\mathbb{E}[\epsilon_j^2] = v_j$. The conditional mean therefore depends on $B = \alpha + \beta \geq 0$. Put $M = I - B$, which is a nonsingular M-matrix iff $\rho(B) < 1$ [4]. The deterministic skeleton is

$$\dot{K} = -\delta K + \Phi K + \psi(v), \quad (2)$$

$$\dot{v} = \omega + \Gamma K - Mv, \quad (3)$$

$$\dot{x}_i = x_i (f_i - \bar{f}), \quad f_i = A(K_i) - \eta v_i, \quad (4)$$

where $\Phi \geq 0$ is the spillover matrix with $\rho(\Phi) < \delta$, $\Gamma = \text{diag}(\gamma_i)$, and A is a concave benchmark-performance function. Capital growth from compute and data access, $g(K_i, x_i)$ is omitted from (2) for simplicity.

2 Capital–volatility block

Take $n = 1$ and $M = m > 0$, without spillover. The equilibrium satisfies $K^* = \psi(v^*)/\delta$ and $h(v^*) = 0$, where

$$h(v) = mv - \frac{\gamma \psi(v)}{\delta} - \omega. \quad (5)$$

The Jacobian is

$$J = \begin{pmatrix} -\delta & \psi'(v^*) \\ \gamma & -m \end{pmatrix}, \quad \text{tr } J = -(\delta + m) < 0, \quad \det J = \delta m - \gamma \psi'(v^*). \quad (6)$$

Proposition 1. *Let $\Theta = \gamma \psi'(v^*)/(\delta m)$. The equilibrium is locally asymptotically stable iff $\Theta < 1$, and a saddle iff $\Theta > 1$. Moreover $h'(v^*) = m(1 - \Theta)$, so stable equilibria are exactly the upward zero crossings of h .*

Proof. The characteristic polynomial is $\lambda^2 + (\delta + m)\lambda + \det J$, whose discriminant $(\delta - m)^2 + 4\gamma\psi'$ is nonnegative. Both eigenvalues are therefore real, the trace is negative, and stability is decided by the sign of $\det J$ [6, 7]. $\square$

Since the eigenvalues are always real, this block has no Hopf bifurcation. Oscillations would require delays or feedback through (4), which we leave open.

Proposition 2 (Saddle-node to runaway). *If ψ is concave there is a unique positive equilibrium. If ψ is convex and superlinear then h is concave. For small γ there are two equilibria, one stable and one unstable. As γ increases they merge when $\max h = 0$ (where $\Theta = 1$) and then disappear, after which trajectories diverge (an “innovation avalanche”) [7].*

3 Network version

Linearizing (2)–(3) at an equilibrium, with $A_\delta = \delta I - \Phi$ and $\Psi' = \text{diag}(\psi'(v_i^*))$, gives the Metzler matrix

$$J = \begin{pmatrix} -A_\delta & \Psi' \\ \Gamma & -M \end{pmatrix}. \quad (7)$$

It is Hurwitz iff $-J$ is a nonsingular M-matrix [4]. By the Schur complement this holds iff

$$\rho(M^{-1} \Gamma A_\delta^{-1} \Psi') < 1, \quad (8)$$

the matrix form of $\Theta < 1$. A diagonal quadratic Lyapunov function then exists for the perturbation $(\tilde{K}, \tilde{v})$ [5]. Here $A_\delta^{-1} = \sum_k \delta^{-(k+1)} \Phi^k$ amplifies feedback through spillover and $M^{-1} = \sum_k B^k$ amplifies it through volatility persistence.

4 Stochastic stability

With innovation jumps the process (K, v) is a piecewise-deterministic Markov process [8].

Theorem 1. *Let $n = 1$ and $\psi(v) \leq a + bv$ for all v , with $\gamma b < \delta m$. For any c with $b/m < c < \delta/\gamma$, the function $V = K + cv$ satisfies*

$$\mathcal{L}V \leq -\kappa V + C \quad (9)$$

for some $\kappa, C > 0$.

Proof. $\mathcal{L}V = -\delta K + \psi(v) + c(\omega + \gamma K - mv) \leq -(\delta - c\gamma)K - (cm - b)v + c\omega + a$, and both coefficients are positive. $\square$

By Foster–Lyapunov theory [9], together with the Feller property and irreducibility of the jump kernel, the process is positive Harris recurrent with a unique invariant probability measure; existence alone follows from tightness and the Krylov–Bogolyubov argument [10]. The condition $\gamma b < \delta m$ is the global counterpart of $\Theta < 1$ and fails when ψ is superlinear.

5 Selection and monopoly

From (4), exactly,

$$\frac{d}{dt} \log \frac{x_i}{x_j} = f_i - f_j. \quad (10)$$

Proposition 3. *If $f_1 - f_j \geq \Delta > 0$ for all $j \neq 1$ from some time on, then $x_j/x_1 \leq Ce^{-\Delta t}$, so $x_1 \rightarrow 1$ and the entropy $H_x = -\sum_i x_i \log x_i \rightarrow 0$.*

This follows from the replicator identity above [11, 12]. Because f_i depends on K_i , which depends on x_i , replicator dynamics here have no global potential, so we claim no global Lyapunov function for the full (K, v) system. Under timescale separation (fast K, v), (4) reduces to a replicator system with fitness $f_i(x_i) = A(K_i^*(x_i)) - \eta v_i^*(x_i)$, to which the classical theory applies.

Conjecture 1. *There is a critical spillover ϕ_c above which knowledge transfer equalizes fitness and prevents monopoly.*

6 Metrics, calibration, and open problems

Useful metrics are the entropy H_x , the innovation efficiency $I_{\text{eff}} = \Delta \mathbb{E}[A]/\mathbb{E}[v]$, and the systemic risk $v_{\text{sys}} = \sum_i x_i v_i$. Natural empirical proxies are training compute and data scale for K_i , held-out benchmark scores for A_i , cross-run score variance for v_i , and architecture-change frequency for λ_i , in the spirit of scaling-law studies [13, 14]. No data are fitted here. Open problems are: (i) limit cycles with delayed volatility response, (ii) global analysis of (K, v, x) and the value of ϕ_c , (iii) moment bounds when ψ is only mildly superlinear, and (iv) identification of ψ .

References

- [1] P. M. Romer, Endogenous technological change, *J. Political Economy* 98 (1990).
- [2] R. F. Engle, Autoregressive conditional heteroscedasticity, *Econometrica* 50 (1982).
- [3] T. Bollerslev, Generalized autoregressive conditional heteroskedasticity, *J. Econometrics* 31 (1986).
- [4] A. Berman and R. J. Plemmons, *Nonnegative Matrices in the Mathematical Sciences*, SIAM, 1994.
- [5] L. Farina and S. Rinaldi, *Positive Linear Systems*, Wiley, 2000.
- [6] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002.
- [7] Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, 3rd ed., Springer, 2004.
- [8] M. H. A. Davis, Piecewise-deterministic Markov processes, *J. Royal Statistical Society B* 46 (1984).
- [9] S. P. Meyn and R. L. Tweedie, Stability of Markovian processes III: Foster–Lyapunov criteria for continuous-time processes, *Adv. Appl. Probab.* 25 (1993).
- [10] G. Da Prato and J. Zabczyk, *Ergodicity for Infinite Dimensional Systems*, Cambridge University Press, 1996.
- [11] J. Hofbauer and K. Sigmund, *Evolutionary Games and Population Dynamics*, Cambridge University Press, 1998.

- [12] J. W. Weibull, *Evolutionary Game Theory*, MIT Press, 1995.
- [13] J. Kaplan et al., Scaling laws for neural language models, arXiv:2001.08361, 2020.
- [14] J. Hoffmann et al., Training compute-optimal large language models, arXiv:2203.15556, 2022.

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